LESSON 4: Functions Performing Mathematical Calculations

Focus: Applying arithmetic operations, selection and calculations inside functions.

Objective: Use functions to solve mathematical problems and return numeric results.

Exercise 14 – Calculating the Area of a Rectangle

Create a function called **`CalculateArea`**. It must accept the length and width of a rectangle, calculate the area and return the result.

Formula: **`Area = Length × Width`**

Exercise 15 – Calculating the Area of a Circle

Create a function called **`CalculateCircleArea`**. It must accept the radius, calculate the area of a circle and return the result as a Double. Use **`Math.PI`** in your solution.

Formula: Area = π × radius^2

Exercise 16 – Calculating Simple Interest

Create a function called **`CalculateInterest`**. It must accept the principal amount, interest rate and time in years, then calculate and return the simple interest.

Formula: **`Simple Interest = (Principal × Rate × Time) / 100`**

Exercise 17 – Calculating Body Mass Index

Create a function called **`CalculateBMI`**. It must accept a person's weight in kilograms and height in metres, calculate their BMI and return the result. Ensure that the height is greater than zero before performing the calculation.

Formula: **`BMI = Weight / Height²`**